

Northwind Home: grounding the agent in real data and policies

Domain FDE Pre-Work Deep Track • Blue Data Consulting • Stage 6: Ground

Before you start

Read [usecase1_day6_problem_statement.pdf](#) first: it sets out why people need trusted, sourced answers from Northwind's records and policies. This brief is the practical: how an agent is grounded in Google ADK, and how you will prove the grounding works.

The technical question: **does more data make a better answer, or does the shape of the data matter more?** Today you ground the same agent twice, once on raw tables and once on an ontology-aligned layer, and let the score decide.

Today's decision

Which sources the agent may use, in what shape, reached through whom, and how much context each answer is allowed to cost.

Three grounding techniques

Technique	What it does	In the kit
RAG	Finds the policy passages most relevant to a question, and answers from them with a citation	search_policy over six policy files. Local search; Vertex AI RAG as an option
NLP2SQL	Turns a question into one read-only SQL query, runs it, and answers from the rows	run_sql and describe_data. SQLite; BigQuery as an option
MCP	A standard way to reach tools another team runs. The owner decides what is exposed	One Northwind server offers all three tools. The agents connect to it, and never touch the data directly

Same records, two shapes

	Raw tables	Semantic layer (ontology-aligned views)
Names	cmp_tbl, rtn_tbl, vf, grd, dl	complaints_v, returns_v, customer_is_vulnerable
Grades	Codes 1 to 3, with no key	A, B or C, with what each means
Vulnerability	On the customer, and on some accounts. They disagree	Read from the customer, as the Day 4 ontology says
Joins	The agent must work them out	Already done in each view

Three agents, one golden set

Agent	Sees	Role in the experiment
v6_ungrounded	Nothing but its instruction	Before: what the model knows on its own
v6_grounded_raw	The raw tables and the policies, over MCP	Run 1
v6_grounded_semantic	The views and the policies, over MCP	Run 2. Only the shape of the data changed

Ten golden questions: five need records, three need policy, two need both. Every answer must come from a source.

What was built

One MCP server owns the database and the policies. The Day 4 ontology decides what the semantic layer calls things.

	Day 5	Day 6
Knowledge	The instruction and one record per lookup	Records by SQL and policies by search, reached on demand
Data access	Tools that read the data directly	An MCP server that owns the data and decides what is exposed
Safety	Rules on actions	A read-only SQL guardrail: one SELECT, allowed tables only, at most 50 rows
Proof	An ungrounded baseline	A before and after on 10 golden questions, with tokens counted per answer

Your task: 50 minutes, in pairs

CXM and SCM pairs. Work in usecase01_day6. Every command is in README-training-account.md. Record everything in Day6_Worksheets_UC01.docx. Do not change the data, views or policies; propose changes in ADR-6.

#	Part	Time	What you do
1	Predict it cold	10 min	For each golden question: which source answers it, and will raw or semantic get it right?
2	Look inside	10 min	Run the component tests. Compare what describe_data shows in raw and semantic mode.
3	Run it	15 min	Ask the three agents the same questions in the chat, then run the before and after on all ten.
4	Prove it	15 min	Diagnose three failures as context, tool or reasoning. Set a context budget, and write ADR-6.

Seven questions

1. Which golden questions did the semantic layer fix? For each, what in the raw data led the agent astray?
2. Did the semantic agent use more tokens or fewer than the raw one? Was the extra context worth it?
3. Pick one failure. Was it context, a tool, or reasoning? What evidence decided it?
4. In the raw tables, vulnerability appears twice: vf and vuln. Which did your raw agent use, and how would a person know which to trust?
5. A policy answer quotes the delay policy. How do you know it came from the document, not the model?
6. Billing owns the refund ledger. What does MCP give Billing that a shared database password does not?
7. What context budget would you set per answer, and what would you do when a question needs more?

Bring to the debrief

The golden question the shape of the data changed · the failure you could not diagnose · the one column or policy you would rewrite first.

All data is made up. Nothing changes in any real system. If the environment breaks, hand up.